Are There Any Alternate Binding Pockets on The RAC1 Surface?

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Abstract

RAC1 (Ras-related C3 botulinum toxin substrate 1) is a molecular switch catalysing by hydrolysing GTP to GDP. Owing to the slow nature of GTP hydrolysis and even slower dissociation of GDP, RAC1 is always assisted by Guanine Exchange Factors (GEFs) and GTPase-activating proteins (GAPs). As a result, GEFs and GAPs control the switching mechanism in RAC1. This regulatory control mechanism by several GEFs and GAPS provides a means to modulate the RAC1 activity in several diseases in which it has been implicated. The work described here attempts to understand various dynamical aspects of RAC1 in presence of GTP, GDP and guanine exchange factors (GEFs). The primary objective is to enhance the scope of therapeutic interventions in RAC1-associated disease pathways by locating alternative binding site(s) in the RAC1 surface not observed in X-ray derived tertiary structure. We are aware that targeting the nucleotide binding pocket is challenging due to the picomolar binding affinity and high levels of GTP to compete with the drug molecules. we embarked on finding alternative pockets on RAC1 that can modulate its activity. We observe that the active conformation of RAC1, GTP- or GEF-bound, opens a cryptic pocket that lies below the switch I region. This pocket

could be targeted by small molecules to control the flexibility of switch I, which is crucial for nucleotide exchange, and therefore, for GEF binding.

Introduction

Small GTPases, a ubiquitous class of proteins, play pivotal roles in various cellular processes. Their functions include signal transduction, vesicle trafficking, cytoskeletal dynamics, and cell division. Ras GTPases regulate cell proliferation, differentiation, and survival. While Rho GTPases govern cytoskeletal dynamics and cell migration. Their activity is regulated through a molecular switch mechanism by guanine nucleotide exchange factors (GEFs), GTPase-activating proteins (GAPs), and guanine nucleotide dissociation inhibitors (GDIs).[1] These molecular switches cycle between OFF (GDP-bound inactive) and ON (GTP-bound active) states. This switching mechanism controls various processes like cell growth and differentiation to vesicular and nuclear transport.[1,2] Small GTPases are proteins that possess conserved domains across several families; made up of six β-sheets surrounded by 5 α-helices. All domains are conserved such that they carry a guanine nucleoside binding site divided to recognize the guanine base and phosphate groups and the magnesium ion.[1,3] The switch regions regulate the traffic to the guanine nucleoside binding site. It is also well established that GTPases alone cannot hydrolyse GTP effectively. Therefore, GEFs and GAPs, play a significant role in assisting the switching mechanism of GTPases. They control the dynamics of the switch regions that facilitate either hydrolysis of GTP or dissociation of GDP. GDP:Mg^{2+} dissociates from GTPases very slowly, and therefore, GEFs assist in exchanging GDP for GTP One of the accepted mechanisms of GEFs in the dissociation of GDP is by electrostatic repulsion of magnesium ion that forms strong interaction with GTPases and phosphate groups of GDP. The loss of magnesium ion weakens the affinity of GDP which eventually dissociates. Alternatively, some GEFs can rearrange the switch I or II conformations such that GDP's affinity is reduced facilitating its dissociation. However, GEFs can also combine the two mechanisms to facilitate GDP dissociation.[4,5]

Furthermore, GAPs stabilise the transition state for GTP:Mg^{2+}:GTPases and facilitate the hydrolysis of γ-phosphate group from GTP. taken together, the switching mechanism can be seen as GEFs assist in the GTPase activation while GAPs assist in GTP hydrolysis. Following GTP hydrolysis, the inactive GTPases prevent further signal transduction.[6] In some classes of GTPases, the active and inactive state cycling is conjugated with GTPases being shuttled between the cell membrane and cytosol. Such GTPases are found with post-translational modification to carry long-chain fatty acids at their C-terminal to assist bilayer insertion. GDIs[7,8] sequester the lipid component to prevent membrane insertion to GTPases in the cytosol. This coordinated action of GTPases, GEFs, GAPs, and GDIs is essential for various cellular processes. Dysregulation of these components contributes to pathologies like cancer, neurodegeneration, and inflammatory disorders. For instance, mutations in Ras and its regulators are implicated in oncogenesis, highlighting the therapeutic potential of targeting GTPase signalling pathways.[1] Such mutations are known to disrupt the equilibrium between the active and inactive states of GTPases. It has also been observed that some mutations enhance the affinity of GEFs or GAPs that traps the small GTPases either in an activated state or inactivated state. Small molecule inhibitors targeting GTPase activity or disruptors of GEF/GAP interactions[9] are under development for cancer therapy. However, there are several challenges in targeting GTPases and/or GTPases-GAP/GEF complexes. Firstly, targeting the guanine binding site is challenging owing to the large intracellular concentration of GTP and high affinity.[10] In such cases, finding an allosteric site can avoid such competition with endogenous substrates. GTPase hyper-activation or hyper-inactivation can be corrected by disrupting the protein-protein interaction (PPI) between GTPases and GEFs/GAPs. However, it is known that targeting PPI interfaces is extremely challenging due to shallow cavities.

Methods

- List of PDBS used. All systems modelled are described in Table 1.

Table 1: Detailed system studies

Initial PDB	RAC1 Modulator/LIGAND	Comments
1HE1[11]	GDP:Mg^{2+}	Inactive RAC1
3TH5[12]	GTP:Mg^{2+}	Active RAC1
4YON[13]	PREX1	DH/PH containing GEF
1FOE[14]	TIAM1	DH/PH containing GEF
2VRW[15]	VAV1	DH/PH containing GEF
2YIN[16]	DOCK2	DHR domain containing GEF

Preparation of systems for molecular dynamics simulation:

The coordinates for the protein-ligand complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in **Table 1**. The X-ray structures were imported and processed in the UCSF Chimera v1.17.[17] Any missing loops were built using modeller[18] with UCSF Chimera v1.17 as the GUI.[19,20]

Generating AMBER-compatible parameters:

All systems were prepared for MD simulations with the Leap module in AMBERtools23. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field. Amber-compatible parameters for all nucleotides were obtained from the Manchester AMBER database For the VAV1 zinc finger region the parameters of the Zinc atom, coordinating cysteines and histidine residues, the force field parameters were obtained from the ZAFF[21] force field library.

Molecular dynamics simulations:

All complexes were solvated with TIP3P[22] waters enclosed in a box with dimensions extending 10Å beyond any protein atom, and an appropriate number of neutralizing ions were added to maintain the system's neutrality. The complexes were relaxed using three cycles of minimization of 50000 steps each. The first 5000 steps of minimization adopted the steepest descent algorithm, and the remaining 45000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0Å for the minimization phase. The first cycle of minimization was performed with 25 kcal/(molÅ^2

restraining force on the solute atoms, this force constant was reduced to 5 kcal/(molÅ^2) in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated in 4 stages (see **Table 2**) from 0 K to 300 K for 10 ns with 50 kcal/(molÅ^2 restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps^{-1}. The next step employed the NPT ensemble for density equilibration for another 2 cycles of 10 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 kcal/(molÅ^2. In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 8 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method.[23] The cut-off for evaluating non-bonded interactions was placed at 9.0 Å for the production phase. All the computations were done using the CUDA-enabled PMEMD code[24–26] in AMBER22.[27,28] All simulations were performed with the SHAKE algorithm[29] to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton's equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Table 2: Details of stages of Heating using NVT Ensemble

Stage	**Temperature range (K)**	**Time (ns)**
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Measuring the structural dynamics of RAC1:

Analysis of the MD trajectory was performed with the cpptraj[30] module in AMBER-Tools23. Before the analysis, all solvent atoms and neutralizing ions were stripped off, and the protein conformations were aligned to their respective X-ray conformation. The mass-weighted Cα root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed to understand the domain-specific fluctuations. Furthermore, to understand the specific dynamical changes in RAC1 when bound to different binding partners, we extracted the essential dynamical properties using Principal Component Analysis (PCA). The Variance-Covariance matrix for Cα atoms was extracted in cpptraj[30] module in AMBERTools23. We extracted and analysed 10 principal components, however, the first two principal components explained $< 50\%$ variance in the data. To visualise the essential dynamics, the vectors were projected on the Cα atoms and visualised in VMD 1.9.3.[31]

Identification and measurement of alternate pockets on RAC1:

POVME2[32,33] was employed for identifying various pockets on the RAC1. To be able to incorporate various experimental conformations of RAC1, we employed RAC1 with different binding partners (see **Table 1**). To identify the pockets, the pocketID module on POVME2 was used on aforementioned RAC1 structures. The parameters used for pocket identification are shown in **Table 3**. Following the pocket identification, we monitored the pocket volume changes of these pockets throughout MD trajectories. It has been observed that the pocket might be a good candidate for druggability if its volume range[34–36] was found to lie between 610Å^3 to 930Å^3.

Table 3: Parameters used for pocket identification

Parameter	**Value**
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

Quantifying the GEF binding hotspots on RAC1 surface:

The free energy of binding was computed using the equation (Eq-1) on the conformations saved from the MD simulations using mm-pbsa python script[37] available in AMBER-Tools23;

$$\Delta G_{binding} = \Delta E_{MM} + \Delta H_{GBSA} - T\Delta S \quad (1)$$

Where ΔE_{MM} is the interaction energy between the binding partners computed in the gas phase using a molecular mechanics method. The solvation-free energy or the enthalpy of hydration ΔH_{GBSA} was computed using the Generalized-Born Surface Area (GBSA) method, and the entropic component of binding $T\Delta S$ was not computed due to huge computation cost. The internal dielectric constant for GBSA calculations was set to 1 and the solvent dielectric constant was fixed as 80. For computing free energy of binding using the MM-GBSA method, the OBC's GB model[38] ($igb = 5$) was used. The mbondi3 radii set was employed to define the PB radii for all solute atoms during the preparation of the topology and parameter files in the leap module in AMBERTools23. The ionic strength for GBSA calculations was maintained at 150 mM. All energy components, in an implicit solvent, were calculated using a large cut-off of 999 Å. The surface area for the non-polar solvation term was computed using the linear combination of pairwise overlaps (LCPO) method.[39] To identify the critical amino acid residues that are critical for the protein-protein interactions between RAC1 and GEFs, we computed the pairwise decomposition of the binding affinity value.

Results and discussion

Small GTPases play a significant role in regulating cellular processes owing to their switching mechanism. GTP-bound state is called the *Active state* and the GDP-bound is called the *Inactive state*. The energy obtained from the GTP hydrolysis is used to undergo conformational changes that are partially responsible for the switching mechanism. Since nucleotides bind tightly to the GTPases, their dissociation is assisted by Guanine Exchange Factors (GEFs). There are several GEFs available for this purpose, and specific GEFs active specific GTPases. In case of RAC1, PREX1, TIAM1, VAV1 and DOCK2 are specific GEFs implicated in a wide variety of diseases. Structurally, PREX1, TIAM1 and VAV1 are DH/PH domain containing proteins, while DOCK2 contains DHR domain. Irrespective of their structural diversity, all GEFs bind to the switch I and II regions and assist in the nucleotide exchange, i.e. dissociation of GDP and association of GTP. The nucleotide binding site is well conserved among the GTPase family, and therefore, making it challenging to target for finding small molecule binders. The molar concentrations of GTPs with picomolar levels of binding affinity makes it even more difficult for the competing the small molecules. Therefore, we embarked to understand the complete dynamical landscape of RAC1 protein bound to either nucleotides (GDP or GTP) or GEFs (PREX1, TIAM1, VAV1 or DOCK2). We expect to locate alternate binding pockets, that may improve the druggability of RAC1 protein. We also attempt to understand if different binding parters show differences in the pockets we identify which can be exploited to selectively target RAC1 implicated in different signalling pathways. We hypothesise that the selectivity can be measured as the function of the pocket volume changes. we also attempt to establish if there exists any correlation in the volume changes amongst other pockets or with the nucleotide binding site. The latter can be believed to influence the nucleotide traffic and eventually control the switching mechanism of RAC1.

Rac1 shows possibility of four more binding pockets

We employed the POVME2 tool to identify and measure the pocket volume changes over the course of the MD trajectory. RAC1 was simulated under different conditions (See **Table 1**). Prior to identifying the binding pockets all binding partners including nucleotides were deleted from the trajectory. Pockets were identified using the starting conformation of RAC1 and the pocket volume changes were monitored throughout the course of the simulation. In addition to the nucleotide binding site, we identified four pockets on the RAC1 surface (**Figure 1**). Site P1a (**Figure 1A**) lies are the interjunction between switch I and II, while site P1a1(**Figure 1A**) lies slightly downstream to Site P1a1. These sites forms that interface region where GEFs and GAPs are known to dock and perform their functions. We observer that site P1a is similar to the covalent binder in G12S mutant of K-ras protein.

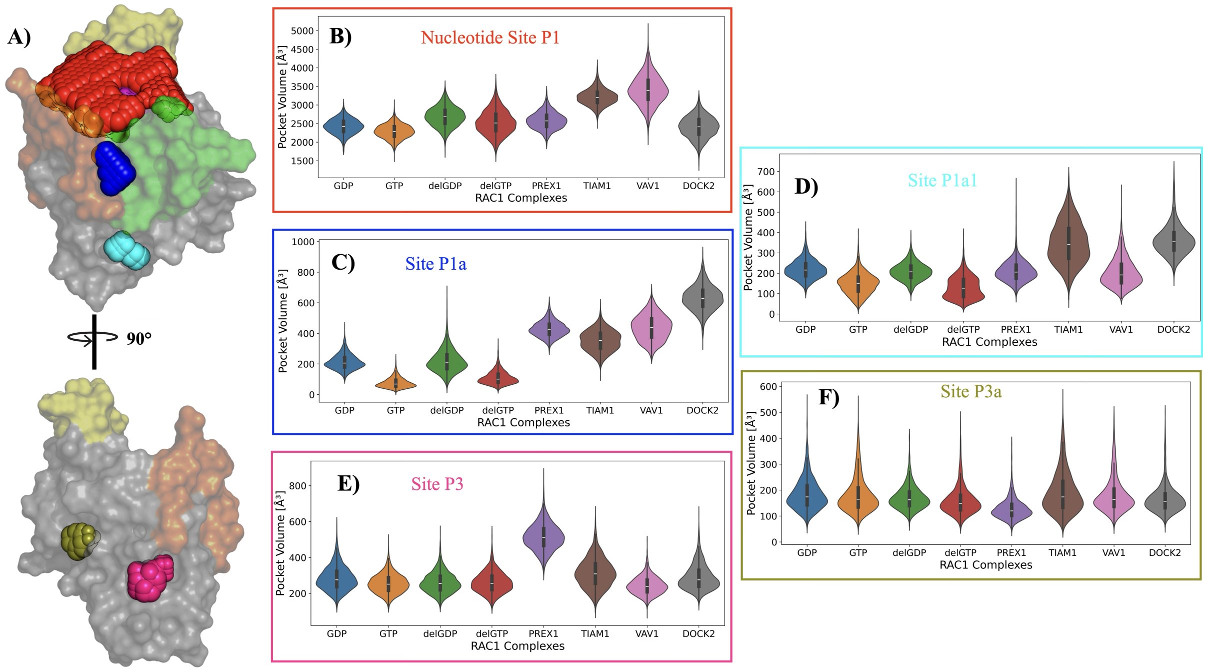

Figure 1: Binding Pockets on RAC1 surface. **A)** Shows the surface model of Rac1 protein with coloured spears illustrating the binding site identified with POVME2. The helical insert is highlighted using the yellow colour. The green region indicates the switch II and orange region indicates the switch I region. The magenta region slightly visible through the red spears indicates the P-loop. The red spears occupy the nucleotide binding site that is lined with helical insert, switch regions and the P-loop forms the phosphate binding region. Blue spears occupy the site P1a and the cyan spears occupy the site P1a1. Site P3 and P3a are visible by rotating the Rac1 by 90°. The salmon pink spears occupy site P3 while the olive green spears occupy site P3a. The violin plots illustrate the distribution of the pocket volumes for each of the binding sites on the Rac1 surface. The changes in the pocket volumes are monitored when Rac1 is complexed with either GEFs, GDP, GTP or in the *apo*-state. **B)** Site P1 or the nucleotide binding site **C)** Site P1a **D)** Site P1a1 **E)** Site P3 and **F)** Site P3a

Nucleotide and GEF binding significantly affects the switch region dynamics

Comments on Structural dynamics

1. Region wise w/ and w/o nucleotides
2. Region wise w/ and w/o GEFs
3. Taken together, comment on 1. and 2.
4. Show with RMSF plots the flexibility of each region

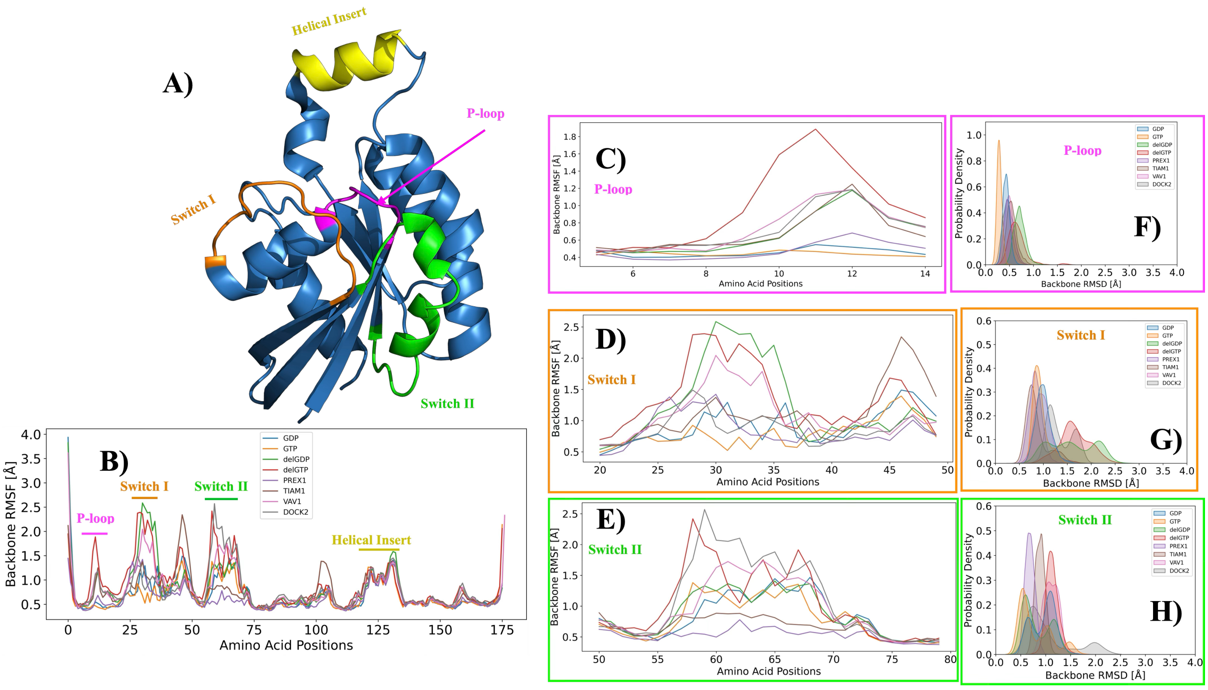

Figure 2: RAC1 domains and dynamics of each region. **A)** A Cartoon representation of human RAC1 (**PDB id:1HE1**). The P-loop region is depicted in the magenta colour. The Switch I region is depicted in Orange Colour, and the Green coloured region is the Switch II region. The helical region is shown in yellow colour. **B)** Atomic fluctuations of the backbone atoms are measured as root-mean-squared fluctuations (RMSF) for human RAC1 domain. **C)** Backbone atom RMSF is shown as the expansion of the P-loop region. **D)** Backbone atom RMSF is shown as the expansion of the Switch I region. **E)** Backbone atom RMSF is shown as the expansion of the Switch II region. **F)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the P-loop. **G)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the Switch I region. **H)** The normalised distribution of the root-mean-squared deviation of the backbone atoms of the Switch II region.delGDP and delGTP are systems from which by deleting the corresponding nucleotides from the RAC1 binding pocket to create an *apo* system.

5. Support with PCA plots for more detailed discussion (later this will support different pocket dynamics)

identifying alternate binding pockets on RAC1 surface

1. Discuss elaborately the RAC1 regions that significantly contribute to GEF binding vis-à-vis X-ray observations on these interactions. a) Link and compare to the dynamical observation of these regions/residues w/o GEFs.

2. Show and discuss the nucleotide site volume changes as follows:

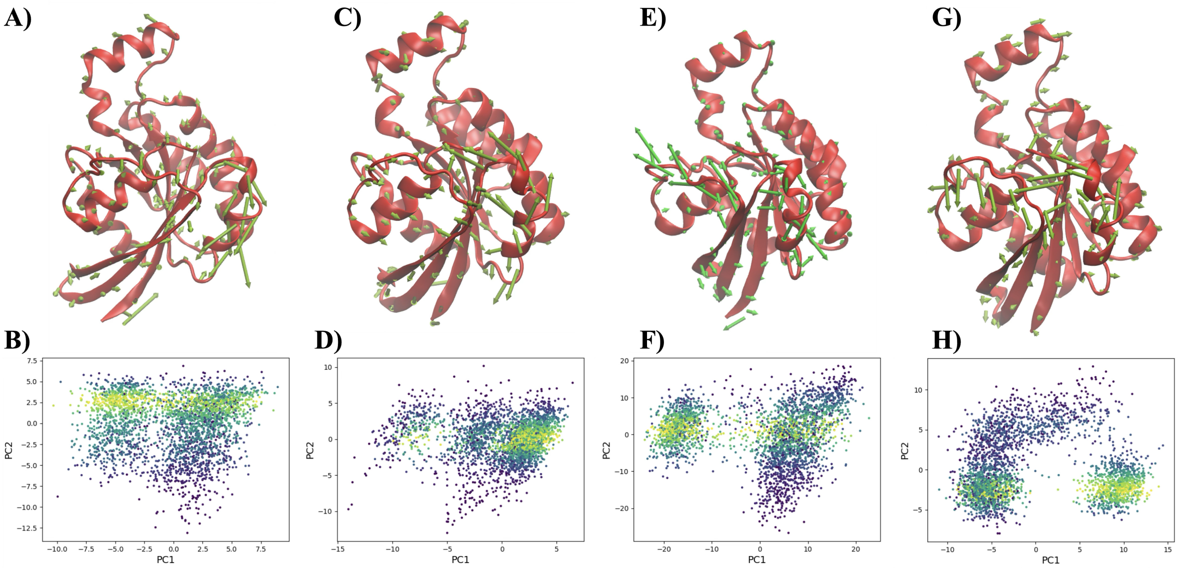

Figure 3: Essential Dynamics of RAC1 protein extracted using principal component analysis.

(a) Inactive (GDP bound) $->$ Effect of GDP deletion

(b) Active (GTP bound) $->$ Effect of GTP deletion

(c) Binding DH/PH containing GEF $->$ PREX1, VAV1 and TIAM1

(d) DHR domain containing GEF $->$ DOCK2

(e) Discuss what is the "Taken Together message"

3. Show and discuss all other pockets

   (a) KRAS-G12C covalent binding pocket $->$ Since this already shown in GTPases (RAS family) elaborate here RAC1 (or RHO family) and discuss if this FIRST time observation or not. If not, then discuss with published work.

   (b) Discuss all other pockets $->$ Find out if any have been reported for RAC1 or RHO family or other GTPases.

4. Link MMGBSA per residue decomposition and contribution of residues surrounding these pockets vis-à-vis different GEFs.

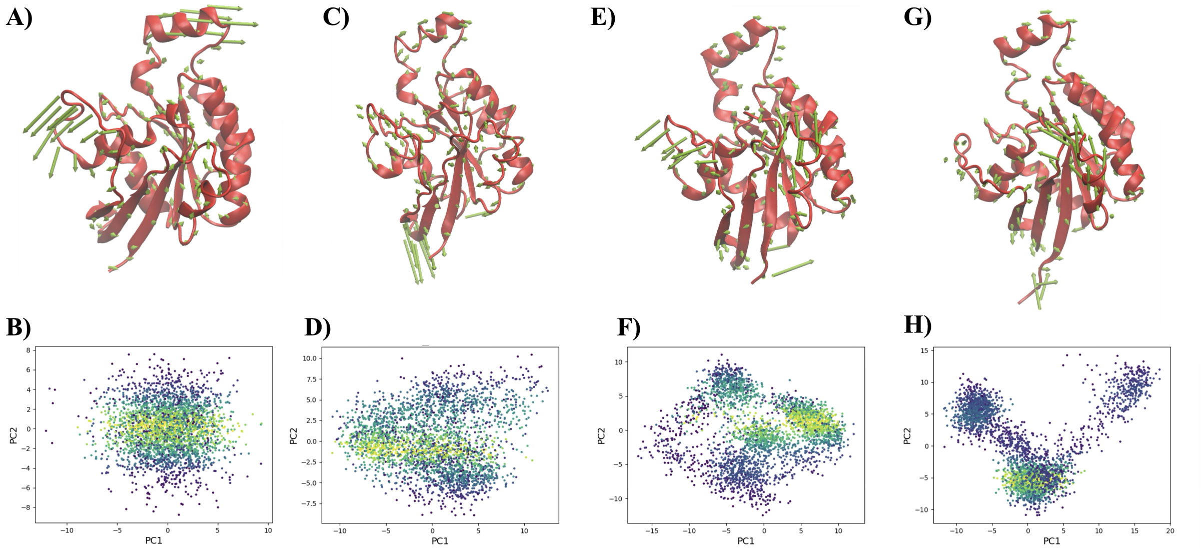

Figure 4: Essential Dynamics of RAC1 protein extracted using principal component analysis.

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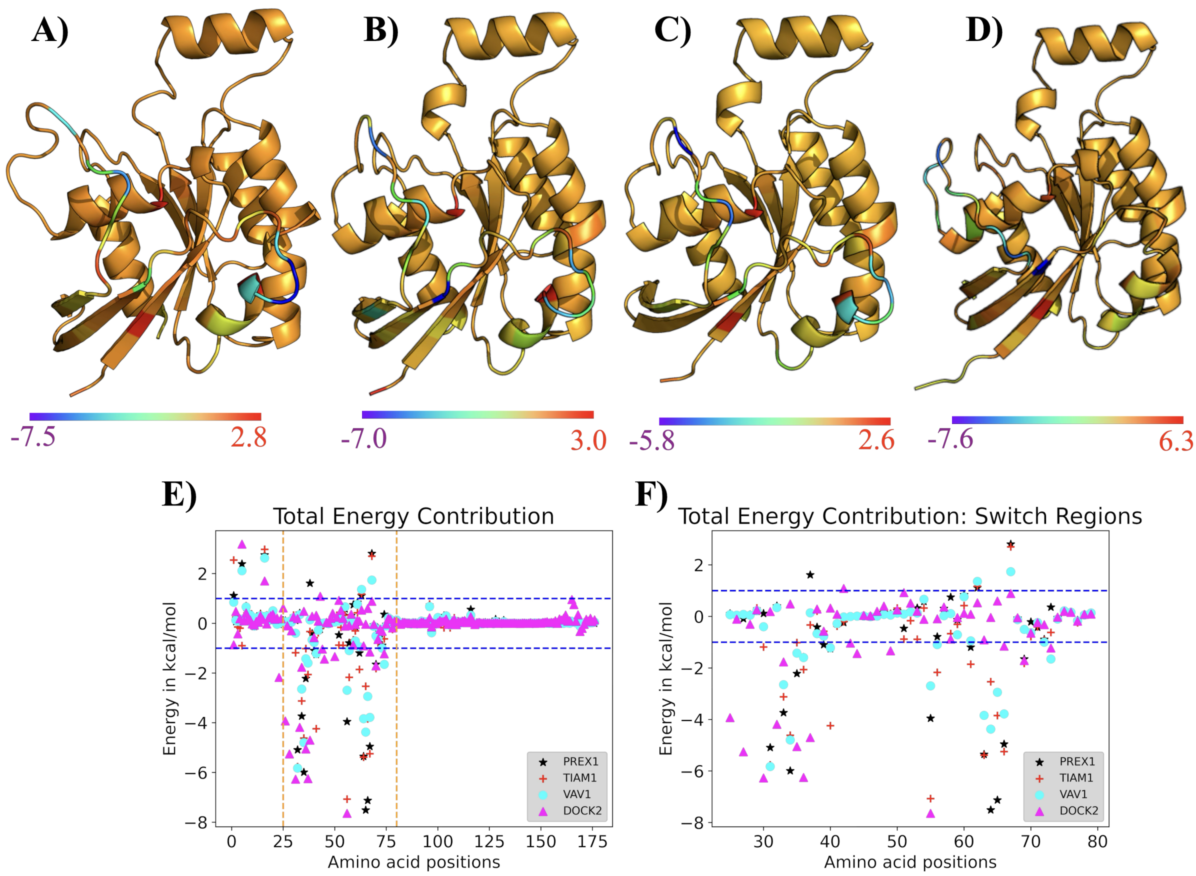

Figure 5: Per Residue contribution from RAC1 protein towards GEF binding.

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